EMPVERIFY

Customer Support Chatbot

Comprehensive Knowledge Base

Prepared by: Agregar Technologies Ltd.

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Part 1: About Agregar Technologies

1.1 Company Overview

Agregar Technologies is a leading Regulatory Technology (RegTech) firm headquartered in Accra, Ghana. Founded in 2022 (formerly known as Gavac Solutions), Agregar Technologies provides AI-powered verification and compliance solutions to businesses across Africa and globally. The company's tagline is "Innovating Trust with AI-Powered Verification."

Agregar Technologies is registered and regulated by the Data Protection Commission of Ghana and is a member of NAPBS (National Association of Professional Background Screeners), USA. The company operates in over 15 countries and has served more than 40 corporate clients, completing over 5,000 verifications.

1.2 Mission and Vision

Mission: To become the leading provider of KYC, KYE, KYV, KYB, and AML solutions in Africa and globally, helping businesses protect assets, reduce liability, and create safer working environments.

Vision: To establish itself as Africa's largest and most reliable RegTech solutions provider through instant, cost-effective verification platforms.

1.3 Leadership Team

Name	Title
Harry Baiden	Founder / CEO
Priscilla Abban	Head, Technology and Product Delivery
Justice Paintsil	Head, Verification
Wendell Neequaye	Head, Commercial
Ebuka Chizota	Lead, Software Engineer
Yaw Owusu	Lead, Data Partnership
Victoria Opoku	Lead, Verification
Sulley Fuseini	Human Resource Lead

1.4 Contact Information

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- **Phone:** +233 (0) 20 984 4520 / +233 (0) 30 250 3018
- **Email:** info@agregartech.com
- **Website:** https://agregartech.com
- **Business Hours:** Monday-Friday, 8:00 AM - 5:00 PM (GMT)

1.5 Compliance and Certifications

- Registered and regulated by the Data Protection Commission of Ghana
- Member of NAPBS (National Association of Professional Background Screeners), USA
- Compliant with FCRA (Fair Credit Reporting Act) - USA
- Compliant with GDPR (General Data Protection Regulation) - EU
- Compliant with local Ghanaian data protection regulations

1.6 Industries Served

Agregar Technologies serves a wide range of industries including:

- Banking and Financial Services
- Healthcare and Pharmaceutical
- Education
- Manufacturing
- E-Commerce and Retail
- Insurance
- Telecommunications
- FMCG and FMCD
- Gaming
- Staffing and Recruitment
- Travel and Hospitality
- Microfinance

1.7 Products and Platforms Overview

Agregar Technologies operates four proprietary platforms:

1. **EmpVerify** - Human capital intelligence platform for pre-employment screening and continuous workforce monitoring.
2. **Autheo** - AI-powered digital identity and compliance platform for regulatory-grade customer onboarding.
3. **Validar** - AI-powered transaction monitoring and AML compliance platform.
4. **Geofy** - Four-dimensional geospatial verification engine combining digital mapping, field operations, GIS intelligence, and geohistory analytics.

1.8 Services Offered

- **KYE (Know Your Employee):** Employee, field agent, and vendor screening before onboarding with continuous risk monitoring.
- **KYC (Know Your Customer):** Customer onboarding and identity verification for regulated industries.
- **KYB (Know Your Business):** Business verification, corporate due diligence, and organizational screening.
- **KYV (Know Your Verifier):** Verification of ground verifiers in the Agregar network.
- **AML Monitoring:** Real-time financial transaction monitoring and anti-money laundering compliance.
- **Vendor and Business Onboarding:** Comprehensive screening for vendor relationships and business partnerships.
- **Continuous Monitoring:** Ongoing fraud detection and compliance monitoring post-onboarding.

Part 2: EmpVerify Platform - Complete Guide

2.1 What is EmpVerify?

EmpVerify is Agregar Technologies' flagship human capital intelligence platform for comprehensive pre-employment background verification and Know Your Business/Customer (KYB/KYC) checks. It is a web-based platform accessible at https://app.empverify.com that enables organizations to conduct thorough verification checks on candidates and businesses before onboarding.

EmpVerify manages the entire background verification workflow - from case creation and candidate consent through document verification, reference checks, and final report generation. It supports multiple verification types, multiple countries, and advanced compliance checks including AML/PEP screening and criminal record verification.

2.2 Key Features

2.2.1 Verification Types

EmpVerify supports the following verification checks, which can be enabled or disabled per case:

Verification Type	Description
Identity Verification	Validates candidate identity using government-issued IDs (Ghana Card, Voter ID, Passport, Driver's License, SSNIT). Includes biometric matching via Smile Identity and NIA cross-validation.
Education Verification	Verifies academic qualifications by contacting educational institutions. Tracks degree type, institution name, enrollment dates, and certificate authenticity.
Employment Verification	Confirms employment history including job titles, dates, responsibilities, and eligibility for rehire. Contacts HR departments and direct supervisors.
Criminal Record Check	Conducts police and criminal background verification. Records criminal history status and police report findings.
Credit Check	Performs credit history verification through data sources like XDS and Kenya credit bureaus. Generates credit reports for financial risk assessment.
AML/Fraud Check	Anti-Money Laundering and fraud screening using SmileID AML API. Checks against PEP lists, sanctions lists, and adverse media.
Sanctions List Screening	Screens candidates against OFAC, UN, EU, and national sanctions lists for regulatory compliance.
Litigation Check	Checks for any ongoing or past litigation involving the candidate.
Address Verification	Verifies residential address through GPS geolocation, physical address confirmation, and utility document verification.
Reference Check	Contacts professional references provided by the candidate. Collects feedback on work quality, character, and professional conduct.
Social Media Screening	AI-powered analysis of candidate social media profiles for inappropriate content, unprofessional behavior, and inconsistencies.

2.2.2 Know Your Business (KYB) Module

EmpVerify includes a full KYB module for business verification. This module allows clients to verify companies and their beneficial ownership structures before engaging in business relationships.

KYB verification levels include:

- **Basic:** Company registration number verification, TIN number check, business address confirmation, and business license validation.
- **Standard:** All Basic checks plus UBO (Ultimate Beneficial Owner) declaration, company profile review, bank account statement verification, and articles of association review.
- **Enhanced:** All Standard checks plus director identity verification, UBO biometric verification, and in-depth corporate due diligence.

The KYB module captures:

- Company name, registration number, TIN number, and date of incorporation
- Legal structure and business industry classification
- Business address and contact details
- Business documentation and licenses
- Director information (name, role, DOB, nationality, ID details)
- UBO details (ownership percentage, country of residence, ID verification)

2.2.3 Document Verification with AI

EmpVerify integrates with Agregar's Document Verification API (powered by AI/ML) for automated document authenticity checks. The API supports:

- **Ghana Card Verification:** ML-based detection of key features (ECOWAS logo, Ghana coat of arms, etc.), expiry date validation, data consistency checks, OCR extraction of name, ID number, nationality, and all card details.
- **Driver's License Verification:** Authenticity validation, feature detection, expiry check, OCR extraction of name, license number, class, and expiry date.
- **Passport Verification:** Passport authenticity check, MRZ (Machine Readable Zone) extraction for name, passport number, nationality, and date of birth.
- **Face Comparison (KYC):** AI-powered facial recognition comparing a selfie against a document photo. Returns match confidence score, face quality assessment, and similarity metrics.
- **Spoof/Liveness Detection:** Detects print attacks (2D photos), replay attacks (digital screens), mask detection, deep fake detection, and performs quality assessment using MobileNetV2 and MTCNN.
- **AML/PEP Compliance Check:** Real-time screening against global watchlists via SmileID API. Returns PEP status, sanctions status, regulatory status, adverse media status, and risk assessment.
- **Unified Document Verification:** A single API endpoint that handles all document types (Ghana Card, Driver's License, Passport) with face comparison and returns a standardized response with overall verdict, confidence score, and risk level.

2.2.4 Smart Capture (CentralKYC)

EmpVerify uses CentralKYC's Smart Capture technology for document and identity capture during the consent form process. Smart Capture is a Next.js-based web application that provides:

- Real-time Ghana Card capture using device camera with auto-detection
- AI-powered quality indicators to guide users for optimal capture
- Selfie capture with face detection utilities for biometric matching
- Mobile-optimized performance for field operations
- TensorFlow.js-based document detection and auto-capture

- Integration with EmpVerify's consent form workflow for seamless candidate onboarding

The Smart Capture component ensures high-quality document images are captured, reducing verification failures and improving the accuracy of OCR and biometric matching.

2.3 User Roles and Access

EmpVerify supports multiple user roles with different levels of access and permissions:

Role	Description	Key Permissions
Administrator	Full platform access for Agregar staff	Manage all cases, assign agents, approve onboarding, manage users, view all reports, audit trail access
Supervisor	Team lead for verification agents	Oversee assigned agents, review cases, quality control, workload management
Agent	Verification specialist	Process assigned cases, conduct verifications, update case status, record findings
Client	Corporate client account	Create cases, invite candidates, view published reports, manage sub-accounts, access audit trail
Individual Partner	Partner organization user	Access cases for their partner institution, limited to partner-specific data
Grand Partner	Top-level partner account	Multi-institution access, can create sub-accounts, broader data visibility

2.3.1 Client Sub-Accounts

Client organizations can create sub-accounts for their team members. Sub-accounts have configurable roles:

- **Role 1 (Admin):** Full access including report viewing, case creation, and sub-account management.
- **Role 2 (Standard):** Can create cases and view reports.
- **Role 3 (Limited):** Restricted access based on administrator configuration.

2.4 Complete Verification Workflow

2.4.1 Client Onboarding

Before a client organization can use EmpVerify, they must complete the onboarding process:

1. The client registers on the EmpVerify platform by providing company details, contact information, and required documentation.
2. The registration creates a user account with "Pending" status.
3. An EmpVerify administrator reviews the onboarding request.
4. The administrator approves, rejects, or holds the request. Upon approval, a system-generated password is emailed to the client.
5. The client can then log in, create cases, and start background verifications.

2.4.2 Case Creation and Candidate Invitation

Once onboarded, clients follow these steps to initiate a background verification:

1. The client creates a new verification case by providing the candidate's name, email, phone number, and selecting the verification checks to be performed.
2. The client selects the candidate's country and identity document type (Ghana Card, Voter ID, Passport, Driver's License, SSNIT, or None).

3. The system validates the candidate's identity document against government databases (GVive API for Ghana documents).
4. A case with a unique ID (e.g., EV-0001, EV-ABC001) is created in "Pending" status.
5. A consent form email is sent to the candidate via SendGrid.

2.4.3 Candidate Consent Form

The candidate receives an email with a link to the consent form. The consent form collects:

- **Personal Information:** Full name, date of birth, email, phone, residential address (with GPS coordinates).
- **Education History:** Institution name, degree type, field of study, enrollment and completion dates, certificate uploads.
- **Employment History:** Company name, job title, start and end dates, responsibilities, reason for leaving, supervisor contact details, rehire eligibility.
- **Professional References:** Referee name, company, position, email, phone, and relationship to candidate.
- **Work Experience:** Additional work history with skills listing.
- **Certifications:** Professional certifications with issuing body, date, and document uploads.
- **Social Media Profiles:** LinkedIn, Twitter, Facebook, Instagram URLs.
- **Digital Signature:** Electronic consent signature for authorization of background checks.

2.4.4 Verification Processing

Once the consent form is submitted, EmpVerify's administrators and agents process the verification:

1. An administrator assigns the case to a verification agent.
2. Identity Verification: The system automatically validates the candidate's ID document via GVive API and Smile Identity. AI-powered document verification checks the authenticity of uploaded documents. Biometric matching compares the candidate's photo with their ID.
3. Education Verification: The agent sends verification request forms to educational institutions (registrar's office). The registrar responds with verification of enrollment dates, degree awarded, and graduation status. The response is recorded and the education record is updated with a verification status (All Correct, Progress, Caution).
4. Employment Verification: The agent contacts former employers' HR departments and direct supervisors. HR/supervisor response forms collect: dates of employment, job title, responsibilities, performance ratings (quality of work, communication, attendance, timekeeping, competency, character, colleague relations), and rehire eligibility. The agent records the verification status.
5. Criminal Record Check: The agent processes police verification and criminal background checks. Records whether any criminal history exists and uploads police reports.
6. Reference Check: The agent sends reference forms to the candidate's professional referees. Referees provide feedback on the candidate's work quality, character, and professional conduct.
7. AML/Sanctions/Credit/Litigation: Automated checks are performed against relevant databases and the results are recorded.
8. Address Verification: Physical address is verified using GPS coordinates and, where applicable, field verification by ground verifiers using the Geofy platform.

2.4.5 Verification Statuses

Each verification check can have the following statuses:

Status	Meaning	Risk Implication
All Correct	Verification completed successfully with all information confirmed accurate.	Low risk - Score multiplier: 1.0
Progress	Verification is in progress or partially completed.	Medium risk - Score multiplier: 0.5
Caution	Discrepancies found or information could not be fully verified.	High risk - Score multiplier: 0.3
Not Verified	Verification not yet started or no response received.	Highest risk - Score multiplier: 0.0

2.4.6 Case Completion and Reporting

Once all selected verification checks are completed:

1. The agent marks the case as "Completed" and it is reviewed by a supervisor.
2. The case is "Published" (made available to the client) by an authorized user.
3. The client receives an email notification that the verification report is ready.
4. The client can view the full verification report on the platform, including all findings, documents, and risk assessment.
5. Reports can be downloaded as PDF documents.

2.5 Supported Countries and ID Types

EmpVerify supports verification across multiple countries. For Ghana specifically, the following government ID types are supported with real-time validation:

ID Type	Validation Method	Data Extracted
Ghana Card (NIA)	GVive API + Smile Identity + NIA Response + AI Document Verification	Full name, ID number, nationality, date of birth, photo, expiry date
Voter ID (Old & New)	GVive API (voter and oldvoter endpoints)	Full name, sex, photo
Passport	GVive API + AI Passport Verification with MRZ extraction	Full name, date of birth, gender, photo, passport number, nationality
Driver's License	GVive API + AI License Verification	Name, date of birth, license class, driver image, expiry date
SSNIT	GVive API (ssnit endpoint)	Full name, birth date, sex, photo

For international candidates, the system supports verification without government ID validation, using document uploads and manual verification processes.

2.6 Email Notifications

EmpVerify sends automated email notifications throughout the verification process using SendGrid:

- Candidate Consent Form Request - Invitation to complete the consent form
- Consent Form Received Confirmation - Acknowledges receipt of candidate's consent form
- Foreign Candidate Consent Form (NIA) - Special form for foreign nationals
- Reference Form Request - Sent to professional referees
- Reference Form Received - Confirms referee response
- Case Completed Notification - Alerts client when verification is done
- Case Published Notification - Informs client that report is available
- Onboarding Acceptance - Welcome email with login credentials

- Onboarding Pending - Status update for pending applications
- Onboarding Rejection - Notification of declined application
- Password Reset - OTP token for password reset
- Dynamic Emails - Template-based emails for KYB consent forms, registrar requests, HR/supervisor verification requests

2.7 Audit Trail

EmpVerify maintains a comprehensive audit trail that logs all significant actions on the platform. Every action is timestamped and attributed to a specific user. Tracked actions include:

- User login and logout events with IP addresses
- Password reset requests and completions
- Client candidate invitations and signup events
- Case initiation and document verification events
- Consent form submissions
- Report viewing events
- Sub-account creation
- Administrator case assignments
- Onboarding approvals and rejections
- Verification module assignments
- All recording of verification results (education, employment, criminal, AML, credit, litigation, address, social media, sanctions, identity)
- Case archival and resend events

2.8 Dispute Resolution

EmpVerify includes a dispute management system that allows candidates or clients to raise disputes about verification findings. The dispute system tracks:

- Case reference and consent form linkage
- Dispute description and supporting evidence
- Dispute status (Pending, Resolved, Rejected)
- Resolution notes and outcome
- Date of creation and resolution

Part 3: Employee Risk Scoring System

3.1 Overview

The Employee Risk Scoring System is an AI-powered FastAPI application that evaluates job candidates by computing a comprehensive risk score out of 100 based on background verification data from EmpVerify and a Hyperledger Fabric blockchain-based ex-employee repository. It uses Google Gemini AI for referee sentiment analysis and social media screening, and produces a clear hiring recommendation.

3.2 How Risk Scoring Works

The scoring engine evaluates 7 weighted components that total to 100 points. Each component has detailed sub-criteria that are analyzed to produce a final score.

3.2.1 Component 1: Identity Verification (15 points)

- **Government-issued ID validity (8 pts):** Evaluates the ID verification status from the consent form. 'All Correct' = full score, 'Progress' = 50%, 'Caution' = 30%.
- **Biometric matching (4 pts):** 2 points for document verification, 1 point for Smile Identity response, 1 point for NIA response.
- **Liveness check (2 pts):** Full points if liveness check was performed and passed.
- **Cross-database verification bonus (1 pt):** Bonus point if both Smile Identity AND NIA responses are present.

3.2.2 Component 2: Address Verification (10 points)

- **Residential address verification (6 pts):** Based on verification status scoring.
- **GPS geolocation (2 pts):** Both GPS latitude and longitude present on consent form.
- **Proof of residence (2 pts):** 1 point for physical address on file, 1 point for digital/GPS address.

3.2.3 Component 3: Employment Verification (20 points)

- **Verification status (8 pts per record):** Maps employment verification status through the standard scoring.
- **Rehire eligibility (3 pts per record):** Eligible for rehire = 3 points, not eligible = 0 points.
- **Contact depth (1.5 pts per record):** 0.5 points each for supervisor name/email, 0.25 points each for HR name/email.
- **Tenure consistency (0.5 pts per record):** Both start and end dates present.
- **Supervisor performance bonus (up to 3 pts):** Based on average of quality_of_work, communication, and attendance ratings from direct supervisors.

Employment records are averaged and normalized to 17 points, plus the supervisor bonus of up to 3 points.

3.2.4 Component 4: Ex-Employee Repository (20 points)

- **Rehire eligibility across employers (6 pts):** Calculated as (rehire_eligible / total_employers) * 6.
- **Supervisor behavioral assessment (10 pts):** Evaluates 7 traits per supervisor: quality_of_work, colleague_relations, communication, attendance, timekeeping, competency, character. Each rated on a scale from excellent (1.0) to poor (0.0).

- **Blockchain records (4 pts):** 2 points for existence of blockchain records (1 point per record, max 2). 2 points for departure reason analysis: clean departures (voluntary/resign/retire) = 2 points, mixed = 1 point, involuntary terminations = 0 points.

3.2.5 Component 5: Criminal Record Check (20 points)

- **Criminal verification status (10 pts):** Based on the criminal verification status from the consent form.
- **Clean criminal history (5 pts):** 5 points if no criminal history, 0 points if criminal history exists.
- **AML/Fraud check (2 pts):** Based on AML/fraud verification status.
- **Sanctions list (2 pts):** Based on sanctions list verification status.
- **Litigation check (1 pt):** Based on litigation check verification status.

3.2.6 Component 6: Education Verification (10 points)

- **Degree verification (5 pts per record):** Based on education verification status.
- **Certificate on file (1.5 pts per record):** Full points if certificate document is uploaded.
- **School/institution response (1.5 pts per record):** Points for institution response received.

Education records are averaged and normalized to 10 points.

3.2.7 Component 7: Referee and Social Media (5 points)

- **Reference verification (2 pts):** Calculated as (verified_references / total_references) * 2.
- **Social media screening status (1 pt):** Based on consent form social media verification status.
- **AI referee sentiment (1 pt):** Google Gemini AI analyzes referee feedback for sentiment. Score normalized from [-1, 1] to [0, 1].
- **AI social media score (1 pt):** AI analysis of social media profiles normalized from 0-100 to 0-1.

3.3 Risk Levels and Hiring Decisions

Score Range	Risk Level	Hiring Decision	Meaning
>= 80	LOW	APPROVE	Candidate passes all checks with high confidence. Recommended for hiring.
60 - 79	MEDIUM	CONDITIONAL_APPROVE	Some areas need attention. May hire with conditions or additional checks.
40 - 59	HIGH	FURTHER_REVIEW	Significant concerns detected. Requires manual review by hiring team.
< 40	CRITICAL	REJECT	Major red flags. Not recommended for hiring.

3.4 Industry-Specific Weight Profiles

The scoring system supports different weight profiles for different industries, reflecting varying regulatory and risk requirements:

Component	Default	Financial	Healthcare	Technology	Government
Identity	15%	15%	15%	12%	20%
Address	10%	8%	10%	8%	10%
Employment	20%	18%	18%	22%	18%
Ex-Employee	20%	17%	15%	18%	17%
Criminal	20%	25%	27%	18%	25%
Education	10%	10%	10%	12%	5%
Referee/Social	5%	7%	5%	10%	5%

3.5 AI-Generated Narrative

The risk scoring system uses Google Gemini Pro AI to generate a 3-5 sentence executive summary for hiring managers. This narrative explains the risk assessment in plain English, highlights key positive and negative factors, and provides actionable insight. If the AI service is unavailable, a template-based narrative is generated as a fallback.

3.6 Confidence Score

Each risk assessment includes a confidence score that reflects the completeness of the data used for scoring. The confidence calculation checks whether identity, criminal, education, and employment checks were performed, and whether education, employment, supervisor, and reference records exist. The formula is: (completeness_ratio * 85%) + 15%, ensuring a minimum confidence of 15%.

3.7 Blockchain Integration

The Employee Risk Scoring System integrates with a Hyperledger Fabric blockchain-based ex-employee repository. This blockchain stores immutable records of employee departures, including:

- Employee name and employer details
- Departure reasons (voluntary resignation, retirement, termination, etc.)
- Rehire eligibility status
- Supervisor behavioral assessments

The blockchain ensures that ex-employee records cannot be tampered with, providing a trusted source of employment history verification. The system searches blockchain records by employee name and employer name, and incorporates the findings into the risk score.

Part 4: Autheo - Digital Identity and Compliance Platform

4.1 Overview

Autheo is Agregar Technologies' AI-powered digital identity and compliance platform designed for regulatory-grade customer onboarding. It provides a comprehensive suite of identity verification tools for organizations that need to comply with KYC regulations.

4.2 Key Features

4.2.1 Document Processing

- Real-time OCR and AI parsing of identity documents (NIN, voter cards, passports, driver's licenses)
- Advanced forgery detection including hologram analysis, font analysis, texture analysis, and tampering detection
- Cross-validation against government and third-party registries

4.2.2 Biometric Verification

- Passive and active liveness detection (blink detection, motion analysis, depth analysis)
- Facial matching with ID photos and government databases
- Optional voiceprint authentication

4.2.3 Video KYC

- Live agent-assisted video verification sessions
- Automated video KYC for self-service onboarding
- Real-time document capture and verification during video calls

4.2.4 Business Identity Verification

- Corporate registration verification
- Ultimate Beneficial Owner (UBO) discovery and verification
- TIN and VAT verification
- Shareholder verification
- Director identity linkage and cross-referencing
- PEP (Politically Exposed Persons) screening
- Sanctions and adverse media screening

4.3 Use Cases

- Remote and in-branch customer onboarding for banks
- Digital wallet and bank account opening
- SIM registration for telecoms
- Fintech app user verification
- ATM and kiosk-based verification

Part 5: Validar - Transaction Monitoring and AML Platform

5.1 Overview

Validar is Agregar Technologies' AI-powered transaction monitoring and Anti-Money Laundering (AML) compliance platform. It provides real-time detection and prevention of financial crimes for digital banks, fintechs, and payment service providers.

5.2 Key Features

5.2.1 Transaction Monitoring

- Rule-based and AI/ML anomaly detection
- Real-time, batch, and streaming data support
- Fraud scenario templates (smurfing, structuring, duplicate accounts)

5.2.2 Name and Sanctions Screening

- Real-time screening against OFAC, UN, EU, and national sanctions lists
- PEP (Politically Exposed Persons) screening
- Adverse media monitoring

5.2.3 Risk Assessment

- Behavioral profiling and customer segmentation
- Dynamic customer risk scoring updated per transaction
- Real-time alerts for Suspicious Transaction Reports (STR) and Currency Transaction Reports (CTR)

5.2.4 Compliance Management

- Automated STR/SAR generation in regulator-ready formats
- Case management system with complete audit trails
- Compliance dashboard for team collaboration
- Regulatory reporting automation

5.3 Use Cases

- AML compliance for digital banks and fintechs
- High-risk industry monitoring (crypto, gambling, forex)
- Cross-border payment screening
- Real-time fraud detection for payment service providers

Part 6: Geofy - Geospatial Verification Engine

6.1 Overview

Geofy is Agregar Technologies' four-dimensional geospatial verification engine that combines digital mapping, field operations, GIS intelligence, and geohistory analytics. It provides comprehensive address and location verification services.

6.2 Key Features

6.2.1 Digital Address Validation

- GIS mapping and satellite triangulation for address verification
- GPS coordinate capture and validation
- Digital address matching and confirmation

6.2.2 Physical Field Verification

- Field agent dispatch with timestamped geotag verification
- Supporting document collection (tenancy agreements, utility bills, property images)
- Geo-tagged photos with structured metadata

6.2.3 Geohistory Analytics

- 90-day to 12-month historical location pattern tracking via mobile or GPS
- Alert triggers for location anomalies and sudden relocations
- Change notification engine with auto re-verification triggers

6.2.4 Property Intelligence

- Property ownership verification and asset tagging (residential, commercial, undeveloped)
- Property value assessment
- Multi-property tracking

6.3 Use Cases

- Loan onboarding address verification for banks and microfinance
- Field staff tracking and route optimization
- KYC/KYB compliance address verification
- Asset leasing verification
- Insurance property verification
- Delivery logistics address confirmation

6.4 Ground Verifiers Program

Agregar operates a field verifier program where individuals conduct physical verification tasks across West Africa using the Geofy platform. Ground verifiers perform:

- Address checks and physical verification visits
- Court verifications for litigation checks
- Guarantor and asset checks
- Police verification facilitation

Requirements for ground verifiers:

- Android phone (OS 6.0 or higher)
- Age 18 or above
- Secondary school certificate minimum
- Working email address
- Dependable transportation

Compensation: Up to 300 Ghanaian cedis daily with flexible scheduling and shift selection.

Part 7: API and Technical Reference

7.1 EmpVerify REST API

EmpVerify provides a comprehensive REST API for programmatic access. All API endpoints require authentication via JWT tokens.

7.1.1 Authentication Endpoints

Endpoint	Method	Description
/api/login/	POST	Authenticate user and receive JWT access token
/api/password-reset/	POST	Request password reset token (6-digit OTP sent via email)
/api/password-reset-confirm/	POST	Confirm password reset with OTP token

7.1.2 Case Management Endpoints

Endpoint	Method	Description
/api/cases/	GET	List all cases accessible to the authenticated user
/api/cases/	POST	Create a new verification case
/api/cases/{code}/	GET	Get case details by case code
/api/cases/{code}/education/	GET	Get education verification records for a case
/api/cases/{code}/employment/	GET	Get employment verification records for a case
/api/cases/{code}/directsupervisor/	GET	Get direct supervisor records for a case
/api/cases/{code}/references/	GET	Get reference records for a case
/api/initiate-case/	POST	Initiate candidate verification with ID validation
/api/candidate-invite/	POST	Send candidate invitation email

7.1.3 KYB API Endpoints

Endpoint	Method	Description
/api/kyb/cases/	GET	List all KYB cases
/api/kyb/cases/	POST	Create a new KYB verification case
/api/kyb/cases/{id}/	GET	Get KYB case details
/api/kyb/consent-form/	POST	Submit KYB consent form response
/api/kyb/cases/{id}/directors/	GET	Get directors for a KYB case
/api/kyb/cases/{id}/ubos/	GET	Get UBOs for a KYB case

7.2 Document Verification API

The Document Verification API is a FastAPI-based service running on AWS that provides AI-powered document verification. All endpoints are prefixed with /api/v1/.

7.2.1 API Endpoints

Endpoint	Method	Description
/api/v1/card/verify-ghana-card-sync	POST	Ghana Card verification and OCR extraction
/api/v1/license/verify-license-complete	POST	Driver's License verification and OCR
/api/v1/passport/verify-passport	GET	Passport verification and MRZ extraction
/api/v1/kyc/compare-faces-sync	POST	Face comparison between two images
/api/v1/aml/check	POST	AML/PEP compliance check via SmileID
/api/v1/spoof/analyze	POST	Spoof/liveness detection analysis
/api/v1/document/verify	POST	Unified document verification (all types)
/api/v1/verification/verify-ghana-card-complete	POST	Complete Ghana Card verification with face match

7.3 Employee Risk Scoring API

The Employee Risk Scoring API is a FastAPI-based service that provides risk assessment for candidates.

7.3.1 API Endpoints

Endpoint	Method	Description
/api/v1/auth/login	POST	JWT authentication (access + refresh tokens)
/api/v1/auth/register	POST	Create user account
/api/v1/cases/	GET	List all EmpVerify verification cases
/api/v1/cases/{case_code}	GET	Get single case details
/api/v1/cases/{case_code}/full	GET	Get complete case data with all sub-records
/api/v1/scoring/score/{case_code}	POST	Run full risk scoring on a case
/api/v1/scoring/result/{record_id}	GET	Retrieve stored scoring result
/api/v1/scoring/history/{case_code}	GET	All historical scores for a case
/api/v1/scoring/batch	POST	Score multiple cases at once
/api/v1/referee/analyze	POST	AI referee sentiment analysis
/api/v1/social-media/analyze	POST	AI social media profile analysis
/api/v1/social-media/analyze/linkedin/pdf	POST	LinkedIn PDF analysis
/api/v1/feedback/submit	POST	Submit hiring outcome feedback
/api/v1/feedback/weights	GET	Get current scoring weights
/api/v1/health/	GET	Health check (app, Redis, EmpVerify, LLM, blockchain)

Part 8: Frequently Asked Questions (FAQ)

8.1 General Questions

Q: What is EmpVerify?

A: EmpVerify is a comprehensive pre-employment background verification platform built by Agregar Technologies. It enables organizations to conduct thorough verification checks on candidates and businesses including identity verification, education verification, employment history, criminal background checks, credit checks, AML/PEP screening, and more. The platform is accessible at https://app.empverify.com.

Q: Who can use EmpVerify?

A: EmpVerify is designed for businesses and organizations of all sizes that need to verify candidates before hiring. It serves industries including banking, healthcare, education, manufacturing, telecommunications, insurance, microfinance, staffing, and more. Any organization that wants to mitigate hiring risk can benefit from EmpVerify.

Q: What countries does EmpVerify support?

A: EmpVerify supports background verification in over 15 countries. For Ghana specifically, it offers real-time government ID validation through the GVive API. For other countries, it supports manual verification processes with document uploads. The platform supports country-specific verification workflows including Kenya credit reports and various international criminal background checks.

Q: How long does a background verification take?

A: The turnaround time depends on the types of verification selected and the responsiveness of third parties (educational institutions, former employers, government agencies). Automated checks like AML/PEP screening and document verification are completed within minutes. Manual verifications such as education and employment checks typically take 3-10 business days depending on institutional response times.

8.2 Account and Access Questions

Q: How do I sign up for EmpVerify?

A: To sign up, visit the EmpVerify platform and submit a registration request with your company details. The request is reviewed by the Agregar Technologies compliance team. Upon approval, you will receive login credentials via email. The onboarding process typically takes 1-2 business days.

Q: I forgot my password. How do I reset it?

A: Click the 'Forgot Password' link on the login page and enter your registered email address. You will receive a 6-digit OTP (One-Time Password) via email. Enter this OTP on the password reset confirmation page along with your new password. If you don't receive the email, check your spam folder or contact support.

Q: Can I create additional user accounts for my team?

A: Yes, client accounts with admin privileges can create sub-accounts for team members. Navigate to the user management section and add team members with appropriate roles (Admin, Standard, or Limited access). Each sub-account receives its own login credentials.

Q: What is the difference between a Client and a Partner account?

A: A Client account is a direct organizational account that can create and manage verification cases. A Partner account (Individual Partner or Grand Partner) is for organizations that work on behalf of other clients or manage multiple institutions. Grand Partners have multi-institution access and can create sub-accounts across organizations.

8.3 Verification Process Questions

Q: How do I start a new background verification?

A: Log in to EmpVerify, navigate to the case creation section, and provide the candidate's name, email, phone number, country, and identity document type. Select the verification checks you want to perform (education, employment, criminal, etc.) and submit. The candidate will receive an email with a consent form link.

Q: What happens after I create a case?

A: After case creation, the candidate receives an automated email inviting them to complete a consent form. The consent form collects their personal details, education history, employment history, references, and other required information. Once the consent form is submitted, the case is assigned to a verification agent who processes each selected verification check.

Q: What is the consent form?

A: The consent form is an online form that candidates must complete to authorize background verification. It collects comprehensive information including personal details, residential address with GPS coordinates, education history with certificate uploads, employment history with supervisor contacts, professional references, certifications, social media profiles, and a digital signature authorizing the checks.

Q: Can I track the status of a verification?

A: Yes, you can track case status in real-time on the EmpVerify dashboard. Cases go through the following statuses: Pending (awaiting consent form), In Progress (verification ongoing), Completed (all checks done), and Published (report available for review). Each individual verification check also has its own status indicator.

Q: What is the risk score?

A: The risk score is an AI-generated numerical score out of 100 that evaluates a candidate's overall risk level based on all verification findings. Scores above 80 indicate LOW risk (APPROVE), 60-79 indicate MEDIUM risk (CONDITIONAL APPROVE), 40-59 indicate HIGH risk (FURTHER REVIEW), and below 40 indicate CRITICAL risk (REJECT). The score considers identity verification, address verification, employment history, criminal records, education, and more.

Q: What does the confidence score mean?

A: The confidence score indicates how complete the data used for risk scoring was. A higher confidence score means more verification checks were completed and more data was available for analysis. The minimum confidence is 15% (when very little data is available) and increases up to 100% when all verification types are completed with full sub-records.

Q: How does the blockchain ex-employee repository work?

A: The blockchain-based ex-employee repository stores immutable records of employee departures on a Hyperledger Fabric network. When a candidate is scored, the system searches the blockchain for records matching the candidate's name and employer history. Blockchain records include departure reasons, rehire eligibility, and supervisor assessments.

Because the data is on a blockchain, it cannot be tampered with, providing a trusted source of historical employment data.

8.4 KYB (Business Verification) Questions

Q: What is KYB verification?

A: KYB (Know Your Business) verification is the process of verifying the legitimacy and details of a business entity. EmpVerify's KYB module allows you to verify company registration, TIN numbers, business addresses, directors, and ultimate beneficial owners (UBOs). It comes in three levels: Basic, Standard, and Enhanced.

Q: What are the KYB verification levels?

A: Basic KYB verifies company registration, TIN, and business address. Standard KYB adds UBO declaration, company profile, bank statement, and articles of association review. Enhanced KYB includes all Standard checks plus director identity verification and UBO biometric verification.

8.5 Document and Identity Questions

Q: What ID types are accepted?

A: For Ghana, EmpVerify accepts Ghana Card (NIA), Voter ID (old and new), Passport, Driver's License, and SSNIT. For other countries, various national ID documents are accepted. The platform also supports verification without government ID for special cases.

Q: How does document verification work?

A: EmpVerify uses AI-powered document verification that analyzes uploaded document images using machine learning. The system checks for authenticity markers (holograms, logos, security features), performs OCR to extract text data, validates expiry dates, checks data consistency, and performs face comparison between the document photo and a selfie. The system also includes anti-spoofing technology to detect printed photos, digital screens, and deepfakes.

Q: What is Smart Capture?

A: Smart Capture is the AI-powered document capture technology integrated into EmpVerify's consent form workflow. It uses the candidate's device camera to capture high-quality images of identity documents with real-time guidance, quality indicators, and auto-detection. It also captures selfies for biometric matching. Smart Capture runs on CentralKYC, a Next.js application with TensorFlow.js for on-device AI processing.

8.6 Technical and Integration Questions

Q: Does EmpVerify have an API?

A: Yes, EmpVerify provides a comprehensive REST API that allows programmatic access to all platform features including case creation, case management, verification results, KYB operations, and more. API access requires JWT authentication. Contact Agregar Technologies for API documentation and access credentials.

Q: What technology does EmpVerify use?

A: EmpVerify is built on Django (Python) with PostgreSQL database, deployed on AWS. It uses SendGrid for email delivery, Cloudinary and AWS S3 for file storage, Smile Identity and GVive for identity verification, and custom AI/ML models (PyTorch, TensorFlow) for

document verification and spoof detection. The risk scoring engine uses FastAPI with Redis caching and Google Gemini AI.

Q: Is my data secure?

A: Yes, EmpVerify implements multiple security measures including JWT token-based authentication, encrypted data storage on AWS S3, HTTPS encryption for all data in transit, role-based access control, comprehensive audit trail logging, and compliance with GDPR, FCRA, and Ghanaian data protection regulations. All sensitive documents are stored securely and access is strictly controlled.

8.7 Pricing and Support Questions

Q: How is EmpVerify priced?

A: EmpVerify pricing is based on the volume and types of verification checks required. Contact the Agregar Technologies commercial team at info@agregartech.com or call +233 (0) 20 984 4520 for a customized pricing quote based on your organization's needs.

Q: How do I get support?

A: For support, you can reach Agregar Technologies through the following channels:

- **Email:** info@agregartech.com
- **Phone:** +233 (0) 20 984 4520 / +233 (0) 30 250 3018
- **Business Hours:** Monday-Friday, 8:00 AM - 5:00 PM (GMT)
- **Address:** Trinity Baptist Church, UPSA Road, Accra, Ghana

Q: Can I dispute a verification finding?

A: Yes, EmpVerify includes a dispute management system. If you or a candidate disagrees with a verification finding, a dispute can be raised through the platform. Disputes are reviewed, investigated, and resolved by the verification team. All dispute actions are tracked in the audit trail.

Part 9: Glossary of Terms

Term	Definition
AML	Anti-Money Laundering - Regulations and procedures to prevent financial crimes.
Autheo	Agregar's AI-powered digital identity and compliance platform for KYC onboarding.
Blockchain	Distributed ledger technology used for immutable ex-employee records via Hyperledger Fabric.
Case	A verification request in EmpVerify, identified by a unique ID (e.g., EV-0001).
Consent Form	Online form completed by candidates authorizing background verification checks.
CentralKYC	Smart Capture technology for AI-powered document and identity capture.
ECOWAS	Economic Community of West African States - regional integration organization.
EmpVerify	Agregar's flagship background verification platform.
FCRA	Fair Credit Reporting Act - US regulation governing consumer reporting.
GDPR	General Data Protection Regulation - EU data privacy regulation.
Geofy	Agregar's geospatial verification engine for address and location verification.
Ghana Card	Ghana's national identification card issued by NIA.
Ground Verifier	Field agent who performs physical address and document verification.
GVive	Government ID validation API service for Ghanaian documents.
Hyperledger Fabric	Enterprise blockchain framework used for ex-employee repository.
JWT	JSON Web Token - authentication standard used for API access.
KYB	Know Your Business - business entity verification process.
KYC	Know Your Customer - customer identity verification process.
KYE	Know Your Employee - employee screening and verification.
KYV	Know Your Verifier - verification of ground verification agents.
Liveness Check	Anti-spoofing technology that verifies a person is physically present.
MRZ	Machine Readable Zone - coded text area on passports and travel documents.
NAPBS	National Association of Professional Background Screeners.
NIA	National Identification Authority of Ghana.
OCR	Optical Character Recognition - technology for extracting text from images.
PEP	Politically Exposed Person - individual with prominent public functions.
Risk Score	AI-generated numerical score (0-100) evaluating candidate risk level.
Smile Identity	Third-party biometric verification service provider.
Smart Capture	AI-powered document and selfie capture technology in CentralKYC.
SSNIT	Social Security and National Insurance Trust of Ghana.
STR/SAR	Suspicious Transaction Report / Suspicious Activity Report.
UBO	Ultimate Beneficial Owner - person with significant control of a business.
Validar	Agregar's transaction monitoring and AML compliance platform.
XDS	Credit bureau data source for credit verification.

Part 10: Chatbot Quick Response Templates

10.1 Greeting and Welcome

Template: "Hello! Welcome to EmpVerify by Agregar Technologies. I'm here to help you with background verification, KYC/KYB checks, employee risk scoring, and more. How can I assist you today?"

10.2 Product Inquiry Responses

About EmpVerify

Template: "EmpVerify is our comprehensive pre-employment background verification platform. It supports identity verification, education checks, employment history verification, criminal background checks, AML/PEP screening, credit checks, address verification, reference checks, and more. You can access it at https://app.empverify.com. Would you like to know more about a specific feature?"

About Autheo

Template: "Autheo is our AI-powered digital identity platform for regulatory-grade customer onboarding. It features real-time document OCR, advanced forgery detection, facial matching, liveness detection, video KYC, and business identity verification. It's perfect for banks, fintechs, and telecoms. Would you like to schedule a demo?"

About Validar

Template: "Validar is our AI-powered transaction monitoring and AML compliance platform. It provides real-time fraud detection, sanctions screening, behavioral profiling, automated STR/SAR generation, and comprehensive case management. It's designed for digital banks, fintechs, and payment providers. Interested in learning more?"

About Geofy

Template: "Geofy is our geospatial verification engine that combines digital mapping, field verification, GIS intelligence, and location history analytics. It's used for address verification, property intelligence, and field operations management. Would you like details on how it works?"

10.3 Common Support Responses

Password Reset

Template: "To reset your password, go to the EmpVerify login page and click 'Forgot Password.' Enter your registered email address and you'll receive a 6-digit code via email. Use this code to set a new password. If you don't receive the email within 5 minutes, please check your spam folder or contact us at info@agregartech.com."

Case Status Inquiry

Template: "You can check your case status by logging into EmpVerify at https://app.empverify.com. Navigate to your dashboard to see all your cases. Each case shows its current status: Pending (awaiting candidate response), In Progress (being verified), Completed (all checks done), or Published (report available). For specific case inquiries, please provide your case ID (e.g., EV-0001) and I'll look into it."

Verification Timeline

Template: "Verification timelines vary by check type. Automated checks (AML screening, document verification) complete within minutes. Manual verifications (education, employment) typically take 3-10 business days depending on institutional response times. You'll receive an email notification when your report is ready. You can also track progress in real-time on your dashboard."

Technical Support

Template: "For technical issues, please provide your account email, a description of the issue, and any error messages you're seeing. Our support team is available Monday-Friday, 8 AM - 5 PM GMT at info@agregartech.com or +233 (0) 20 984 4520. For urgent issues, please call us directly."

Pricing Inquiry

Template: "EmpVerify pricing is customized based on your verification volume and the types of checks you need. Our commercial team can provide a tailored quote for your organization. Please contact Wendell Neequaye, Head of Commercial, at info@agregartech.com or call +233 (0) 20 984 4520 to discuss your requirements."

10.4 Escalation Response

Template: "I understand your concern. Let me connect you with our support team for personalized assistance. You can reach them at info@agregartech.com or call +233 (0) 20 984 4520 during business hours (Mon-Fri, 8 AM - 5 PM GMT). Alternatively, I can log a support ticket for you and our team will follow up within 24 hours. What would you prefer?"